

Lec 2: Phonetics

(آواشناسی)

(واج شناسی)

Phonemes and phones

Phoneme

- The smallest meaningful contrastive unit in the phonology of a language
- Each language uses small set of phonemes, much smaller than the number of sounds than can be produced by a human
- The number of phonemes varies per language, with most languages having 20-40 phonemes
 - General American has ~40 phonemes (24 consonants, 16 vowels)
 - The Rotokas language (Paupa New Guinea) has ~11 phonemes
 - The Taa language (Botswana) has ~112 phonemes

Phonetic notation

- International Phonetic Alphabet (IPA): consists of about 75 consonants, 25 vowels, and 50 diacritics (to modify base phones)
- TIMIT corpus: uses 61 phones, represented as ASCII characters for machine readability
 - TIMIT only covers English, whereas IPA covers most languages

Phones in the TIMIT Database

TIMIT	IPA	Example	TIMIT	IPA	Example
pcl	p̚	(p closure)	bcl	b̚	(b closure)
tcl	t̚	(t closure)	dcl	d̚	(d closure)
kcl	k̚	(k closure)	gcl	g̚	(g closure)
p	p	pea	b	b	bee
t	t	tea	d	d	day
k	k	key	g	g	gay
q	ʔ	bat	dx	r	dirty
ch	tʃ	choke	jh	dʒ	joke
f	f	fish	v	v	vote
th	θ	thin	dh	ð	then
s	s	sound	z	z	zoo
sh	ʃ	shout	zh	ʒ	azure
m	m	moon	n	n	noon
em	m̩	bottom	en	n̩	button
ng	ŋ	sing	eng	ŋ̩	Washington
nx	ɹ̥	winner	el	l̥	bottle
l	l	like	r	r	right
w	w	wire	y	j	yes
hh	h	hay	hv	ɦ	ahead
er	ɜ̥	bird	axr	ɤ̥	butter
iy	i	beet	ih	ɪ	bit
ey	e	bait	eh	ɛ	bet
ae	æ	bat	aa	ɑ	father
ao	ɔ	bought	ah	ʌ	but
ow	o	boat	uh	ʊ	book
uw	u	boot	ux	ü	toot
aw	ɑ ^w	about	ay	ɑ ^y	bite
oy	ɔ ^y	boy	ax-h	ɤ̥	suspect
ax	ə	about	ix	ɪ	debit
epi		(epenthetic sil.)	pau		(pause)
h#		(silence)			

Phone (واج-آوا)

- The physical sound produced when a phoneme is articulated
- Since the vocal tract can have an infinite number of configurations, there is an infinite number of phones that correspond to a phoneme

Allophone

- A class of phones corresponding to a specific variant of a phoneme
 - Example: aspirated [p^h] and unaspirated [p] in the words *pit* and *spit*
 - Example: /t/ sounds in the words *tub*, *stub*, *but*, *butter*

Coarticulation (هم تلفظی)

- The phenomenon whereby the articulatory configuration of a phoneme is affected by that of its neighboring phonemes
 - As a result, crisp boundaries between phonemes are hard to define
- Coarticulation is involved in the transformation of phonemes into allophones

Branches of phonetics

Phonology vs. phonetics

- Phonology is concerned with the distribution and patterns of speech sounds in a particular language, or in languages in general
- Phonetics is concerned with the study of speech sounds and their production, classification, and transcription
 - In a nutshell, phonetics deals with the physical nature of speech sounds, and not with their relations to other speech sounds in particular languages

Three basic approaches to the study of phonetics

- Articulatory phonetics is concerned with the position, shape and movements of speech articulators
- Acoustic phonetics is concerned with the spectro-temporal properties of the speech sound waves
- Auditory phonetics is concerned with the perception, categorization, and recognition of speech sounds and the role of the auditory system

Articulatory phonetics

- ▶ The first task of articulatory phonetics is to describe speech sound in the terms of position of the vocal organs

Articulatory phonetics

- ▶ نحوه تلفظ آواها و اجزای موثر در تولید گفتار
- ▶ دلیل بررسی: موثر در شکل سیگنال گفتار و بهتر شدن تشخیص در برچسب زنی
- ▶ واحدهای پایه (اجرها) برای بازشناسی گفتار
- ▶ دسته بندی آواها
- ▶ واکه ها (vowel) : آواهای مصوت
- ▶ در حین بیان انسداد قابل توجهی در مسیر تلفظ (زبان و دهان) اتفاق نمی افتد
- ▶ شکل سیگنال : شبه پریودیک
- ▶ همخوانها (consonants): آواهای بی صدا
- ▶ رخداد انسداد یا انقباض در گلو و دهان هنگام تلفظ

Articulatory phonetics

▶ دسته بندی دیگر اواها

▶ **Voiced** (واکدار) : تارهای صوتی بطور متناوب در تولیدشان باز و بسته می شوند

▶ شکل سیگنال : شبه پریودیک

▶ **Unvoiced** (بیواک) : تارهای صوتی در حین تلفظ در مسیر حرکت متناوب ندارند یا از هم دورند

Alternatively, phoneme classes can be divided into

- Continuant: produced by a fixed vocal tract configuration
 - Includes vowels, fricatives, and nasals
- Non-continuant: vocal tract configuration changes over time
 - Diphthongs, semivowels, stops and affricatives

Articulatory phonetics (consonants)

In terms of articulators, consonants can be described by

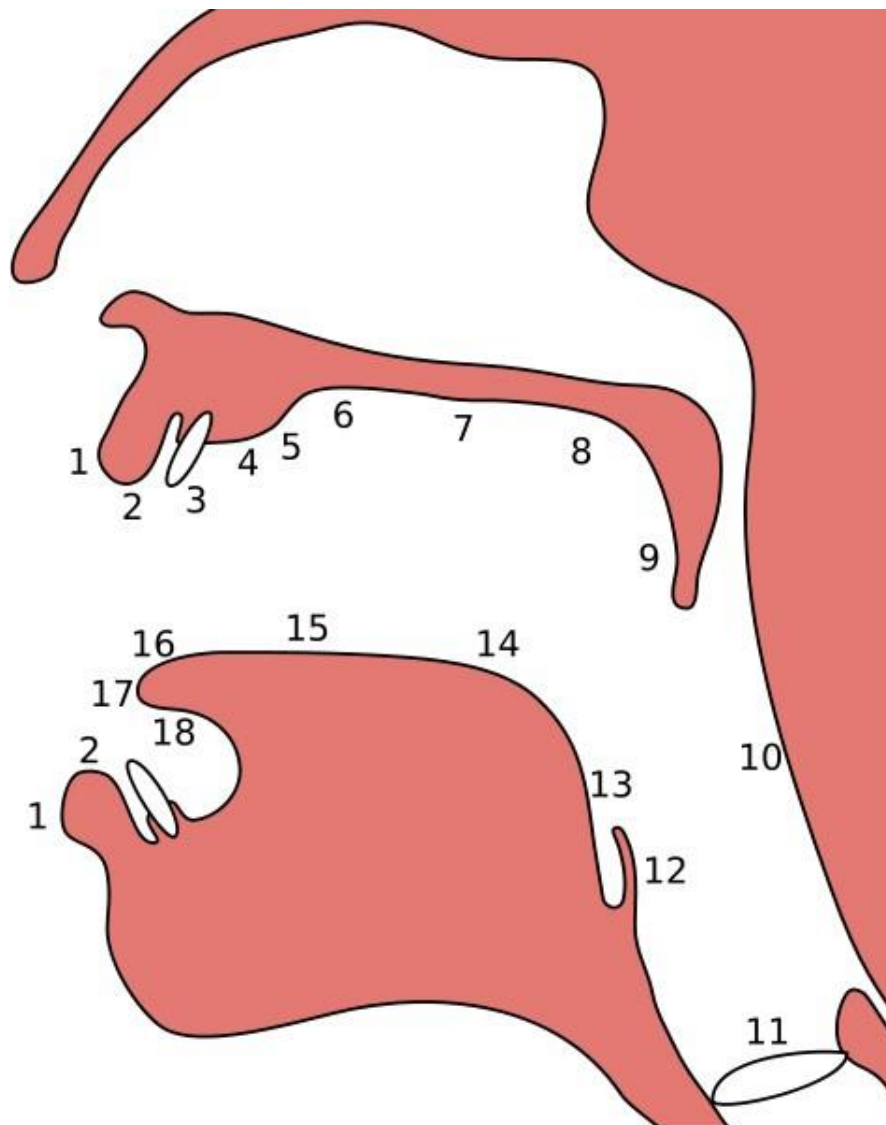
- Place of articulation: Defines the place of contact between an active articulator (i.e. tongue) and a passive articulator (i.e. palate)
- Manner of articulation: Concerned with airflow, the path it takes and the degree to which it is impeded
- Voicing: Determined by the behavior of the vocal folds (vibrating vs. open)

		MANNER	VOICING	PLACE						
				Bilabial	Labiodental	Interdental	Alveolar	Palatal	Velar	Glottal
OBSTRUENTS	Stop		Voiceless	p			t		k	ʔ
			Voiced	b			d		g	
	Fricative		Voiceless		f	θ	s	ʃ		h
			Voiced		v	ð	z	ʒ		
	Affricate		Voiceless					tʃ		
			Voiced					dʒ		
SONORANTS	Nasal		Voiced	m			n		ŋ	
	LIQUID	Lateral	Voiced				l			
		Rhotic	Voiced					r		
	Glide		Voiced	w				j	w	

Place of articulation

- Bilabial: constriction at the lips: [b], [m]
- Labiodental: Lower lips against upper teeth: [f], [v]
- Interdental: constriction between the teeth: [θ] *thing*, [ð] *that*
- Alveolar: constriction is at the alveolar ridge: [t], [n], [z]
- Palatal-alveolar: constriction slightly behind alveolar ridge: [ʃ] *sherry*, [ʒ] *measure*
- Palatal: constriction in the hard palate: [j] *joke*
- Velar: constriction closer to the soft palate: [k], [ŋ] *sing*
- Labiovelar: constriction both at lips and velum: [w]
- Glottal: when closure occurs as far back as the glottis: glottal stop [ʔ], as in the negative utterance uh-uh
- Uvular: constriction in the uvula; none in English, French /r/ in *rouge*

<http://www.uiowa.edu/~acadtech/phonetics/#>



Places of articulation

1. Exo-labial
2. Endo-labial
3. Dental
4. Alveolar
5. Post-alveolar
6. Pre-palatal
7. Palatal
8. Velar
9. Uvular
10. Pharyngeal
11. Glottal
12. Epiglottal
13. Radical
14. Postero-dorsal
15. Antero-dorsal
16. Laminar
17. Apical
18. Sub-apical

Manner of articulation

- Stops: produced by complete stoppage of the airstream: [p]
- Fricatives: tongue comes very close to a full closure: [f], [sh] *sherry*
- Affricatives: combination b/w stops and fricatives: *cherry*
- Nasals: closed oral passage (as in stops), open nasal cavity: [n], [ng] *sing*
- Approximants : halfway between consonants and vowels
 - Liquids: [l], [r]
 - Glides: [y], [w]

Voicing

- When the vocal folds vibrate, it is voiced; otherwise it is voiceless
- Examples of voiceless/voiced: *Sue* vs. *zoo*, *pat* vs. *bat*

<http://www.uiowa.edu/~acadtech/phonetics/#>

Vowels (واکھ ھا)

- The largest phoneme group and most interesting one
 - Carry little information in written speech, but most ASR systems rely heavily on them for performance
- Vowels are voiced (except when whispered) and have the greatest intensity and duration in the range of 50 to 400ms
- Vowels are distinguished mainly by their first three formants
 - However, there is a significant individual variability, so other cues can be employed for discrimination (upper formants, bandwidths)

Diphthongs

- Diphthongs consist of a dynamic vowel sound in which the tongue and lips move between two vowel positions
- Three diphthongs are universally accepted
 - These can be described as a vowel followed by a glide, though the articulators rarely reach the full constriction of a glide
 - Examples: [ay] *bite*, [oy] *boy*, [aw] *bout*
-

Glides and liquids

- Very similar to vowels: periodic, intense waveforms: most energy in low formant; hence, they are also known as semivowels
 - However, they are weaker than vowels since they require a greater constriction of the vocal tract
 - They are also transitional sounds, so their characteristics are strongly influenced by context
- Liquids ([l], [r]) tend to color preceding vowels with their sound
- Glides ([y], [w]) often lengthen vowels to create diphthongs

Nasals (خیشومی ها)

- Produced with glottal excitation (i.e., voicing)
- Vocal tract totally constricted at some point in the oral cavity
 - Thus, the mouth serves as a resonant cavity that traps energy at certain acoustic frequencies, which appear as anti-resonances (zeros of the transfer function)
- Velum is lowered, so air flows through the nasal cavity
 - Since nasal cavity has large area, formant bandwidths are generally broader than for other sonorants
- Nasal consonants are distinguished by their place of constriction
 - [m] at the lips, [n] at the back of the teeth, [ŋ] forward of the velum

Fricatives (سایشی ها)

- Characterized by a major place of constriction
- Come in two flavors: voiced and unvoiced
 - Place of constriction is identical for corresponding pairs of phones
- Unvoiced
 - Vocal tract is excited by a steady air flow, which becomes turbulent at a place of constriction
 - Sorted by place of constriction: [f] *fish*, [θ] *thin*, [s] *sound*, and [ʃ] *shout*
 - Spectrum looks “noisy”
- Voiced
 - Glottis is vibrating, but the airflow becomes turbulent near the constriction, so these sounds display two distinct components
 - Sorted by place of constriction : [v] *vote*, [ð] *then*, [z] *zoo*, [ʒ] *azure*
 - Spectrum contains both “noisy” and harmonic components

Stops (plosives) (انفجاريها)

- Unlike other sounds, which can be described by steady-state spectra, stops are transient phones
 - They are produced by building up pressure behind a constriction and then suddenly releasing it
- As with fricatives, stops come in two flavors: voiced and unvoiced
 - Place of constriction is identical for corresponding pairs of phones: bilabial, alveolar, velar
- Voiced
 - Vocal folds vibrate while pressure builds up, which can be heard as a low-frequency energy radiating through the walls of the throat (voice bar)
 - Sorted: [b] *bee*, [d] *day*, [g] *guy*
- Unvoiced
 - Vocal folds do not vibrate as pressure builds up
 - Sorted: [p] *pea*, [t] *tea*, [k] *key*

Prosody

Long-term variations (over more than one phoneme) in

- Pitch (intonation)
- Amplitude (loudness), and
- Timing (articulation rate or rhythm)

Roles of prosody

- Helps highlight the spoken message
 - Alternation of stressed and unstressed syllables identifies the words that the speaker considers more important
- Helps segment the spoken message
 - Provides cues to syntactic boundaries (e.g., main vs. subordinate clauses) and syntactic structure (e.g., declaratives statements vs. questions)
 - Serves as a “continuity guide” to track speakers in noisy environments
- Provides cues to the state of the speaker
 - F0 and amplitude patterns vary with emotions
- **Interestingly, however, prosody is typically ignored in ASR**

آوانگاری (واج نگاری)

Phonetic notation

- معرفی یک الفبای قراردادی یا استاندارد برای بیان آواهای زبان
- خروجی بازشناس گفتار در سطح آواها، این الفبا است
- تعداد آواها از حروف زبان بیشتر و متنوعتر است
- در برچسب زنی در سطح آواها (واجها) باید این الفبا استفاده شود.

IPA symbol	Arpabet		Examples	IPA symbol	Arpabet		Examples
i	i	IY	heed	v	v	V	verve
ɪ	I	IH	hid	θ	T	TH	thick
e	e	EY	hayed	ð	D	DH	those
ɛ	E	EH	head	s	s	S	cease
æ	(ā)	AE	had	z	z	Z	pizzaz
ɑ	a	AA	hod	ʃ	S	SH	mesh
ɔ	c	AO	hawed	ʒ	Z	ZH	measure
o	o	OW	hoed	h	h	HH	heat
u	U	UH	hood	m	m	M	mom
ʊ	u	UW	who'd	n	n	N	noon
ɜ	R	ER	heard	ŋ	G	NX	ringing
ə	x	AX	<i>a</i> head	l	l	L	lulu
ʌ	A	AH	bud	ɫ	L	EL	battle†
aɪ	Y	AY	hide	m	M	EM	bottom†
aʊ	W	AW	how'd	n	N	EN	button†
ɔɪ	O	OY	boy	f	F	DX	batter‡
ɹ	X	IX	roses	ʔ	Q	Q	§
p	p	P	pop	w	w	W	wow
b	b	B	bob	j	y	Y	yoyo
t	t	T	tug	r	r	R	roar
d	d	D	dug	ʧ	C	CH	church
k	k	K	kick	ʤ	J	JH	judge
g	g	G	gig	ɰ	H	WH	where
f	f	F	fife				

† Vocalic l, m, n ‡ Flapped t § Glottal stop

Some sounds from other European languages (IPA notation):

y	F rue, G Bühne	x	G ich, S México
ʏ	G Hütte	ẽ	F vin
ø	F peu, G Söhne	ã	F vent
œ	F boeuf, G Götter	õ	F vont
ʊ	R ɔl	œ̃	F un

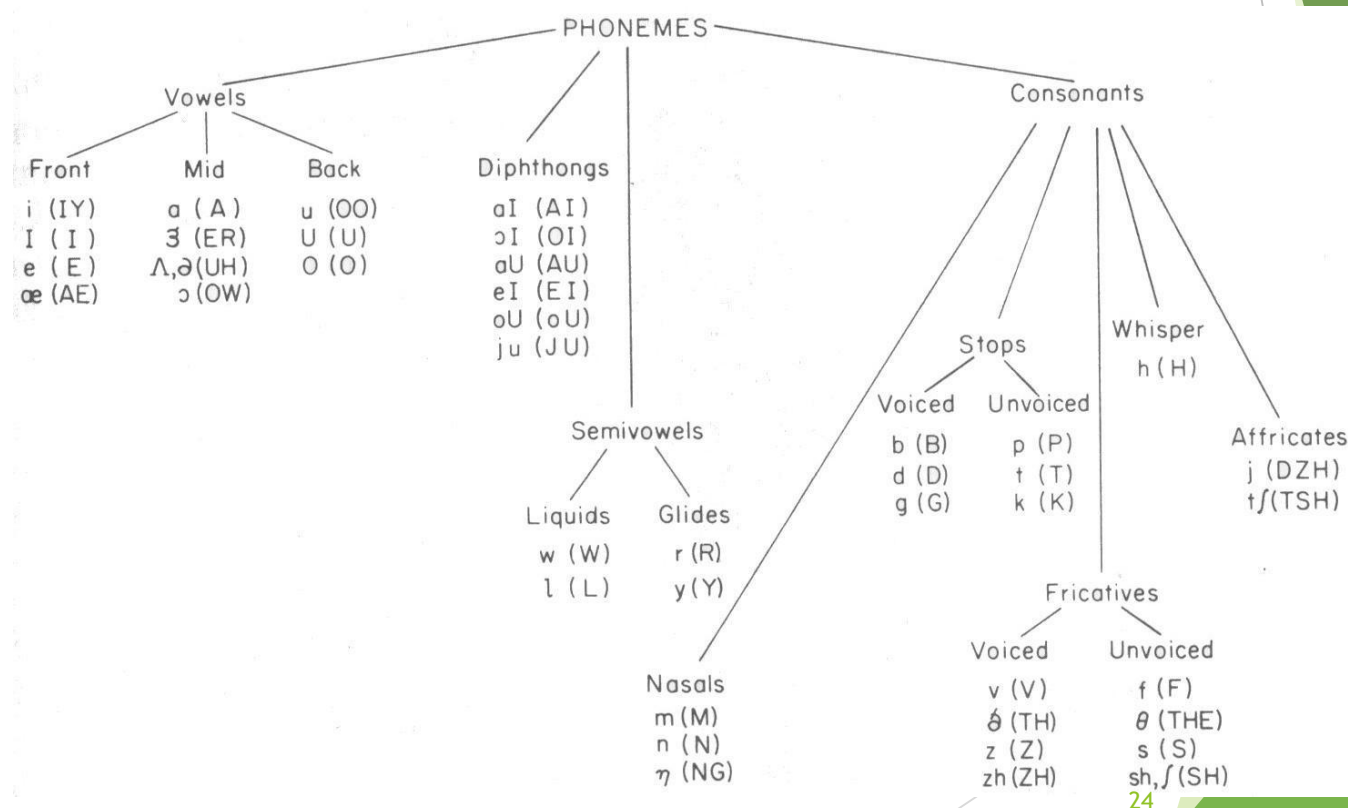
Other marks (IPA):

- : indicates preceding vowel is long
- ˈ precedes an accented syllable

General organization of sounds

Four general classes of sounds in American English

- Vowels, diphthongs, semivowels, and consonants
- Each can be further divided according to articulators (manner, place)



[Rabiner & Schafer, 1978]

Articulatory phonetics (consonants)

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		Voiced					dʒ			
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	LIQUID	Lateral	Voiced				l			
		Rhotic	Voiced					r		
	Glide		Voiced	w				j	w	

واجهای انگلیسی

PHONEME	EXAMPLE	PHONEME	EXAMPLE	PHONEME	EXAMPLE
/iː/	beat	/s/	see	/w/	wet
/ɪ/	bit	/ʃ/	she	/r/	red
/eɪ/	bait	/f/	fee	/l/	let
/ɛ/	bet	/θ/	thief	/y/	yet
/æ/	bat	/z/	z	/m/	meet
/ɑ/	Bob	/ʒ/	Gigi	/n/	neat
/ɔ/	bought	/v/	v	/ŋ/	sing
/ʌ/	but	/ð/	thee	/tʃ/	church
/oʊ/	boat	/p/	pea	/ʃ/	judge
/ʊ/	book	/t/	tea	/h/	heat
/uː/	boot	/k/	key		
/ɜː/	Burt	/b/	bee		
/aɪ/	bite	/d/	Dee		
/ɔɪ/	Boyd	/g/	geese		
/ɑʊ/	bout				
/ə/	about				

○ واکه‌های فارسی

• ۶ واج (+۲ واکه مرکب)

مثال		
علی	ی	i
انسان	ا	e
اکبر	آ	a
بورس	و	u
شد	أ	o
آب	آ	σ
حواله	أو	ou
میل	ای	ei

○ همخوان‌های فارسی

• ۲۵ واج (۲۳ واج)

واج روان / و / در زبانهای عربی و کُردی استفاده می‌شود ولی در فارسی تقریباً فراموش شده است و فقط در بعضی از لهجه‌های فارسی استفاده می‌شود.

شبه واکه	
ل	l
ر	r
م	m
ن	n
ی، یه	j

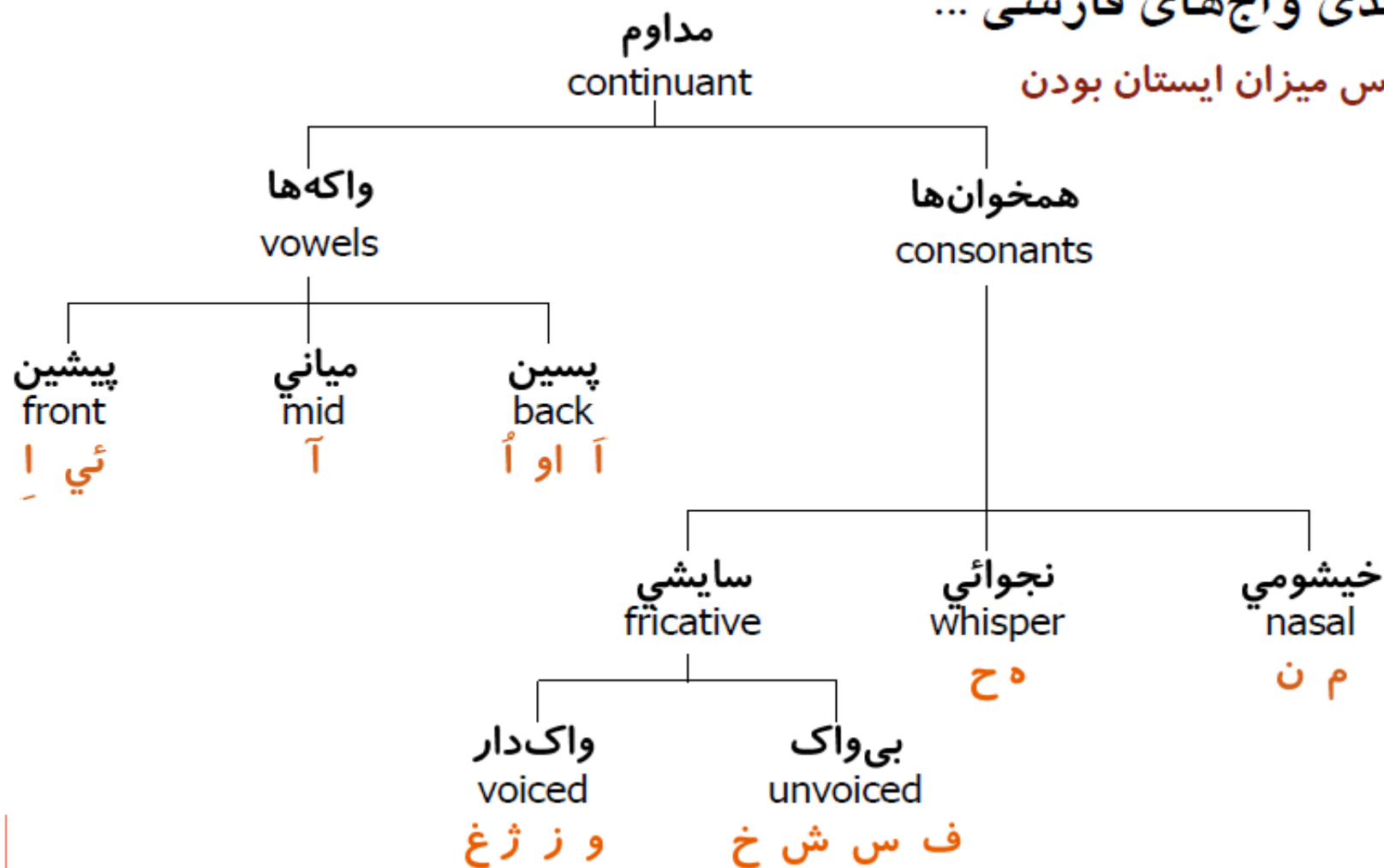
انفجاری-سایشی	
ج	dʒ
چ	tʃ

انفجاری		
	ب	b
	پ	p
	ت، ط	t
	د	d
تاکید	ک (پیشین)	c
کار	ک (پسین)	k
اگر	گ (پیشین)	ɾ
سیگار	گ (پسین)	g
	ق، غ	G
	ع، ء	ɛ

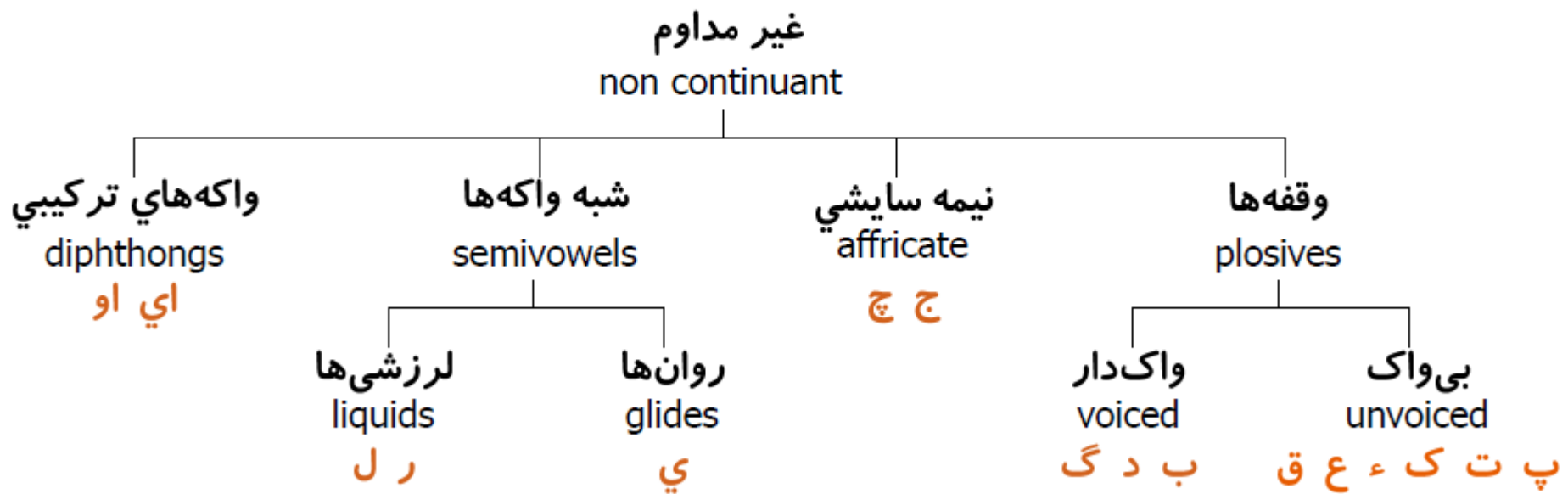
سایشی	
و	v
ف	f
ث، س، ص	s
ز، ذ، ض، ظ	z
ه، ح	h
ش	ʃ
ژ	ʒ
خ	χ

تقسیم‌بندی واج‌های فارسی ...

- بر اساس میزان ایستان بودن



تقسیم‌بندی واج‌های فارسی



هجاها (سیلاب)

c=همخوان
v=واکه

مثال	هجا	نوع
با	CV	۱
تار	CVC	۲
گفت	CVCC	۳

در فارسی: ۳ نوع است

- با واکه شروع نمی‌شوند
 - در نتیجه هیچ واژه‌ای با واکه شروع نمی‌شوند
- حداقل یک و حداکثر ۳ همخوان دارند
- تعداد (با فرض ۶ واکه و ۲۳ همخوان) = بیش‌تر از ۷۶۰۰۰
 - در عمل، حدود ۴۰۰۰
- تعداد هجاها = تعداد واکه‌ها